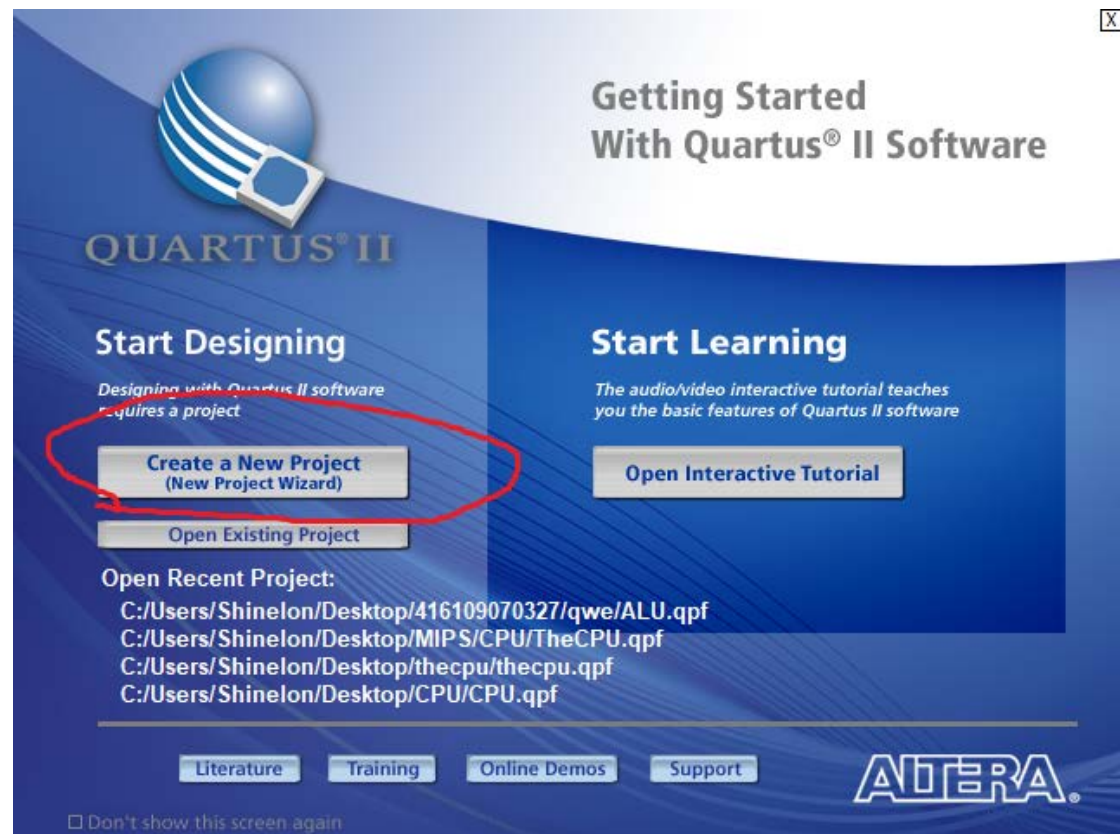


Quartus 使用说明

创建项目

1. 运行 Quartus
2. 选择 Create a New Project



3. 选择项目保存路径，项目名，顶层实体，然后点击 NEXT
注：项目路径不能出现中文，项目名和顶层实体名不能出现中文

New Project Wizard

Directory, Name, Top-Level Entity [page 1 of 5]

What is the working directory for this project?

C:/Users/Shinelon/Desktop/test **项目保存路径**

What is the name of this project?

test **项目名**

What is the name of the top-level design entity for this project? This name is case sensitive and must exactly match the entity name in the design file.

test **顶层实体名**

Use Existing Project Settings...

< Back Next > Finish Cancel Help

4. 选择 NEXT

New Project Wizard

Add Files [page 2 of 5]

Select the design files you want to include in the project. Click Add All to add all design files in the project directory to the project.
Note: you can always add design files to the project later.

File name: ... Add

File Name	Type	Library	Design Entry/Synthesis Tool	HDL Version
-----------	------	---------	-----------------------------	-------------

Add All
Remove
Up
Down
Properties

Specify the path names of any non-default libraries. User Libraries...

< Back Next > Finish Cancel Help

5. 选择器件库，本处选择器件库 Cyclone II，点击 Next

注：该版本尽量选择器件库 Cyclone II，选择其他器件库可能导致无法仿真

The screenshot shows the 'Family & Device Settings' window. The 'Device family' dropdown is set to 'Cyclone II' and is circled in red. The 'Devices' dropdown is set to 'All'. The 'Target device' section has 'Auto device selected by the Fitter' selected. The 'Show in 'Available devices' list' section has 'Package' set to 'Any', 'Pin count' set to 'Any', 'Speed grade' set to 'Any', and 'Name filter' is empty. The 'Show advanced devices' checkbox is checked. The 'Available devices' table lists various Cyclone II devices. The 'Companion device' section has 'HardCopy' set to 'None' and 'Limit DSP & RAM to HardCopy device resources' is unchecked. The 'Next >' button is highlighted.

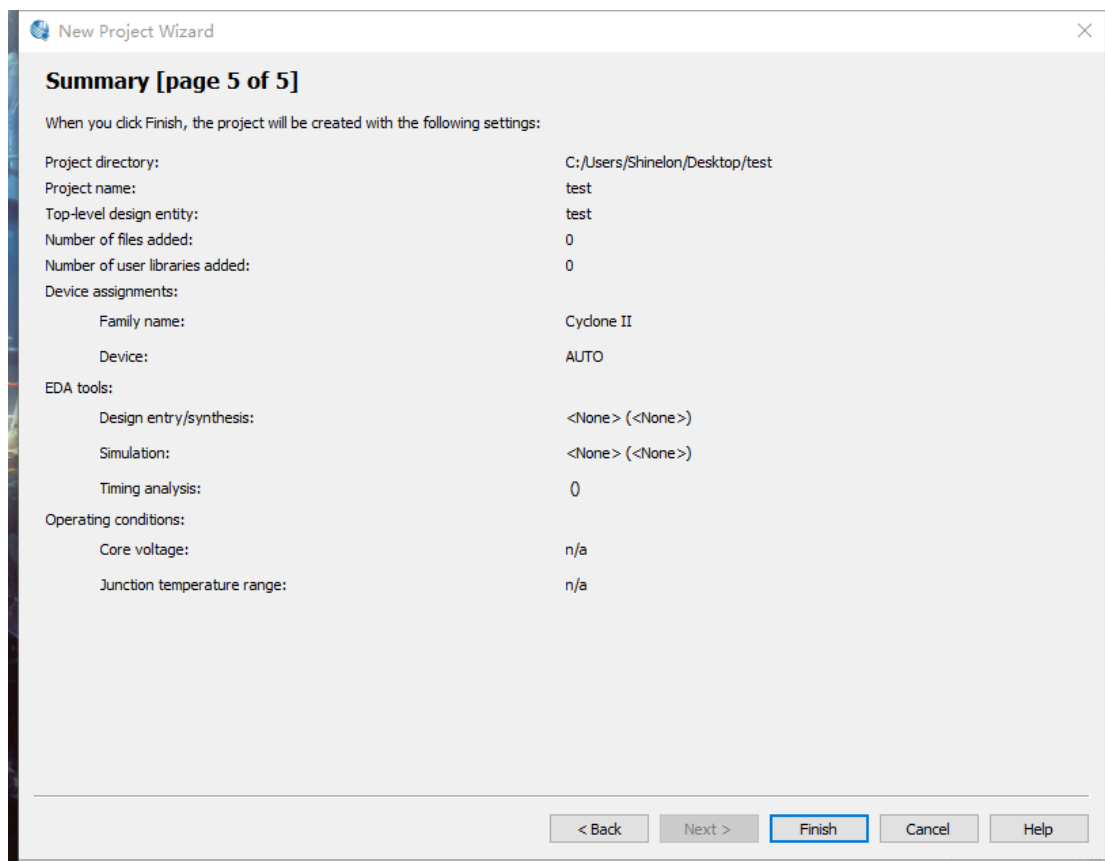
Name	Core Voltage	LEs	User I/Os	Memory Bits	Embedded multiplier 9-bit elements	PLL	
EP2C5AF256A7	1.2V	4608	158	119808	26	2	8
EP2C5AF256I8	1.2V	4608	158	119808	26	2	8
EP2C5AT144A7	1.2V	4608	89	119808	26	2	8
EP2C5F256C6	1.2V	4608	158	119808	26	2	8
EP2C5F256C7	1.2V	4608	158	119808	26	2	8
EP2C5F256C8	1.2V	4608	158	119808	26	2	8
EP2C5F256I8	1.2V	4608	158	119808	26	2	8

6. 选择 NEXT

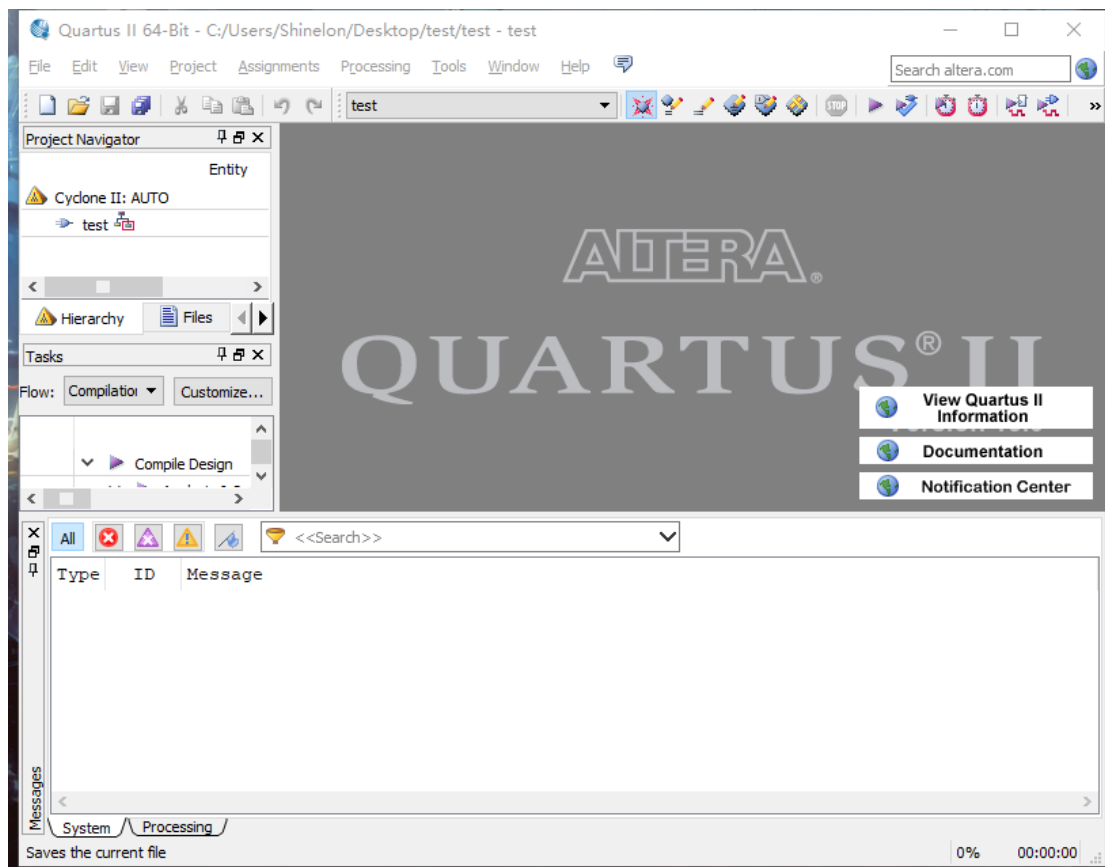
The screenshot shows the 'EDA Tool Settings' window. The 'EDA tools' table has columns for 'Tool Type', 'Tool Name', 'Format(s)', and 'Run Tool Automatically'. The 'Design Entry/Synthesis' row has 'Tool Name' set to '<None>' and 'Format(s)' set to '<None>'. The 'Simulation' row has 'Tool Name' set to '<None>' and 'Format(s)' set to '<None>'. The 'Formal Verification' row has 'Tool Name' set to '<None>' and 'Format(s)' set to '<None>'. The 'Board-Level' row has 'Tool Name' set to 'Timing', 'Format(s)' set to '<None>', and 'Run Tool Automatically' checked. The 'Next >' button is highlighted.

Tool Type	Tool Name	Format(s)	Run Tool Automatically
Design Entry/Synthesis	<None>	<None>	☐ Run this tool automatically to synthesize the current design
Simulation	<None>	<None>	☐ Run gate-level simulation automatically after compilation
Formal Verification	<None>	<None>	
Board-Level	Timing	<None>	☒
	Symbol	<None>	
	Signal Integrity	<None>	
	Boundary Scan	<None>	

7. 选择 Finish

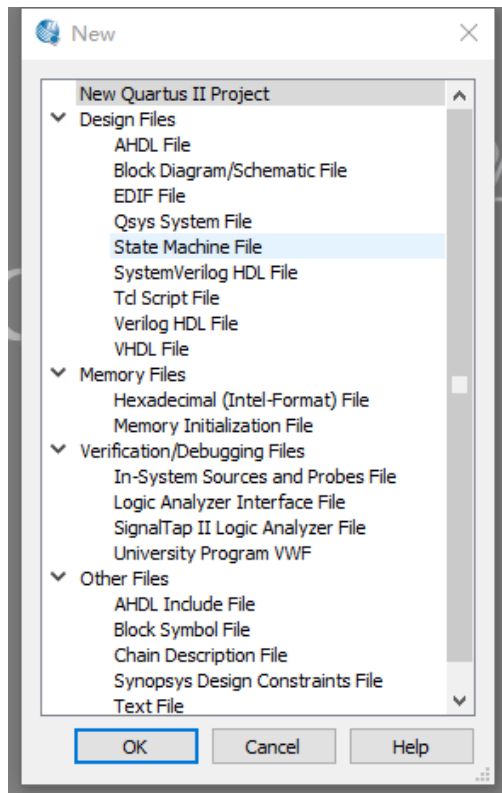


8. 项目创建完成

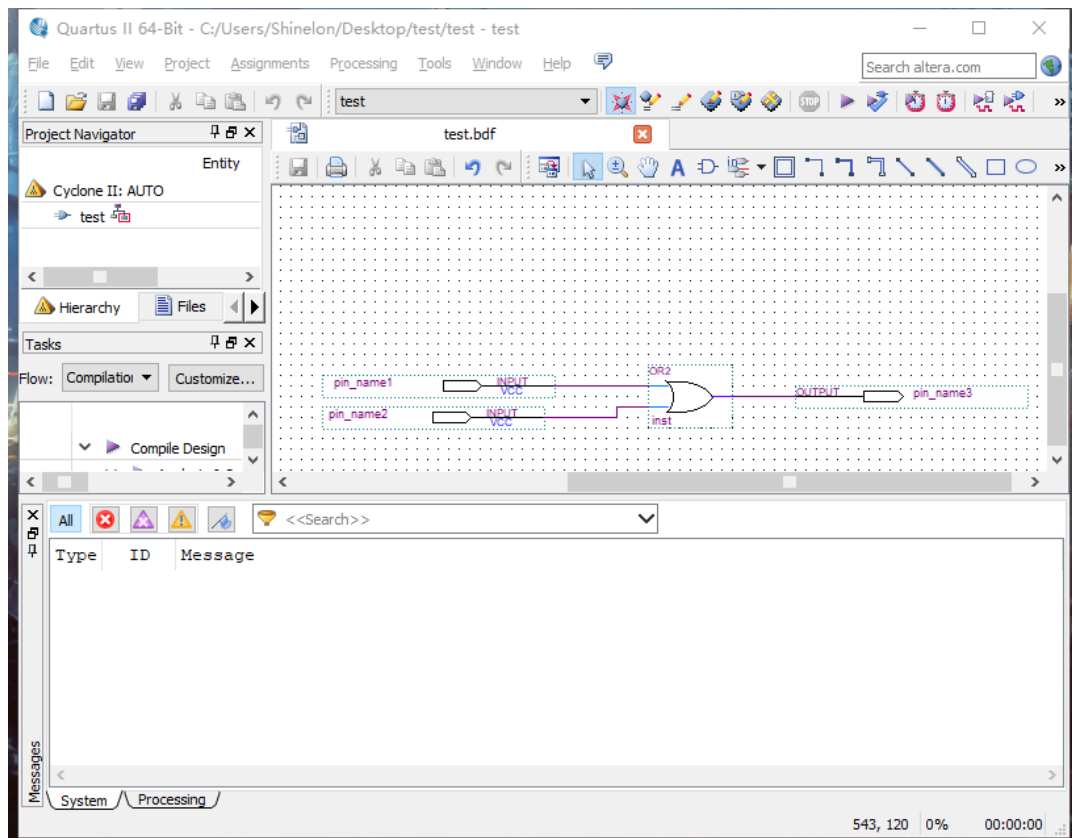


创建电路图文件

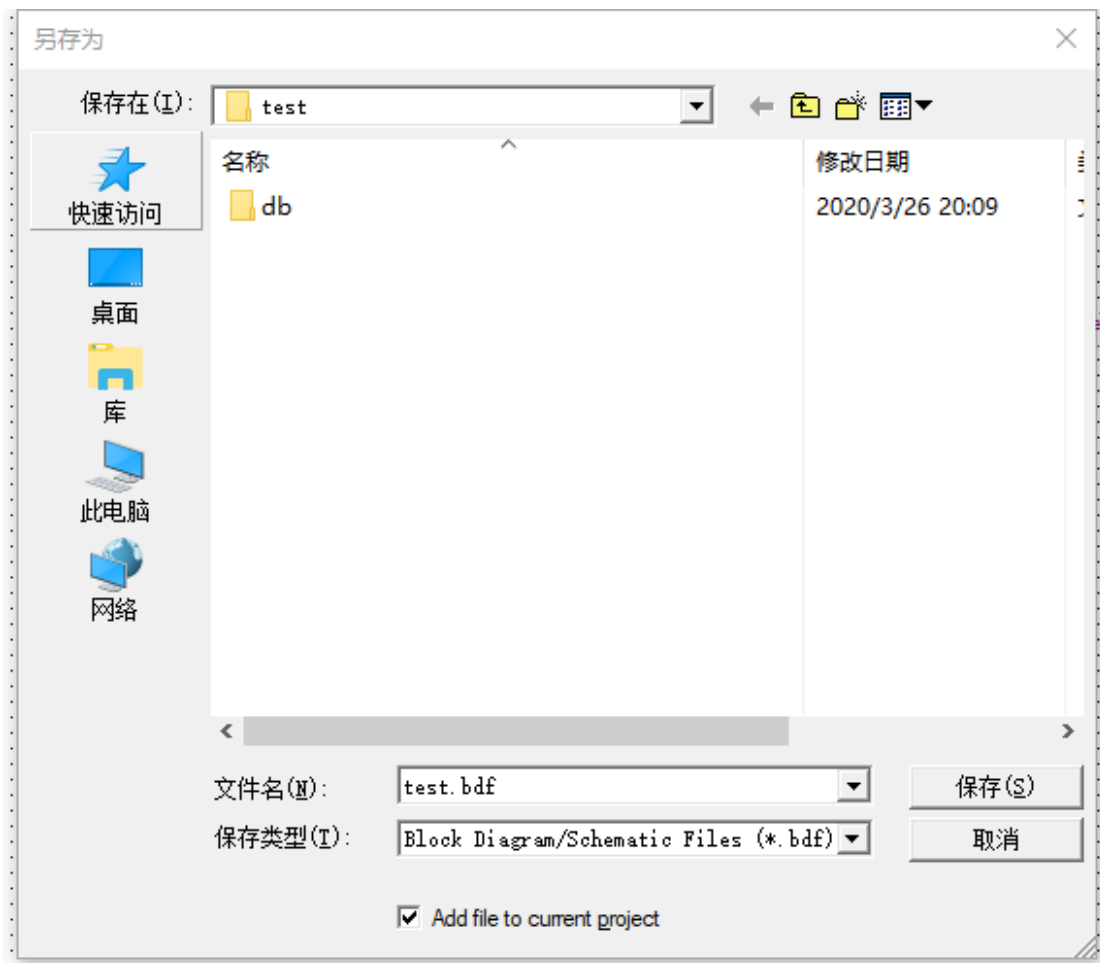
1. 在 File 选项下，选择 New，选择 Block diagram/Schematic File,点击 ok



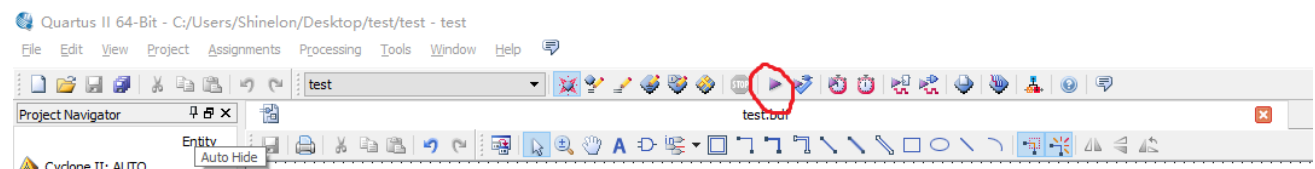
2. 编辑电路图，此处详见课上内容，不做说明



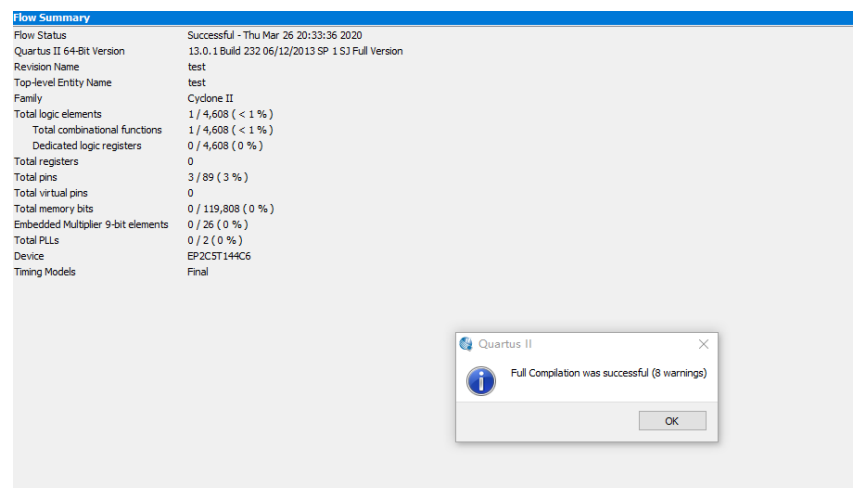
3. 保存文件，单个文件时文件名与顶层实体名相同，文件扩展名为.bdf，选择保存



4. 运行仿真，找到工具栏中红圈按钮（或者红圈后面的按钮），并点击

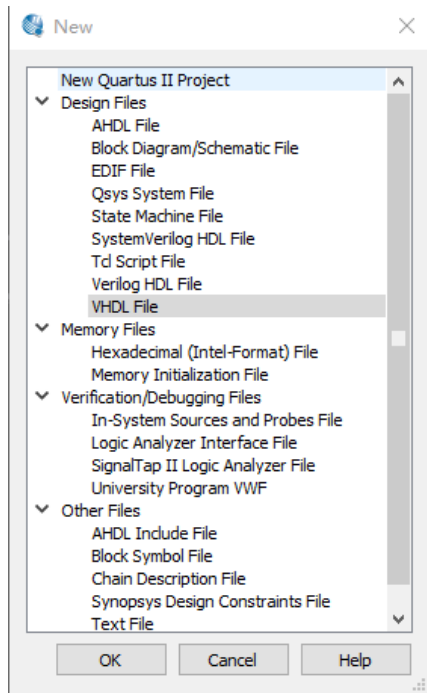


5. 等待仿真完成



创建 VHDL 文件

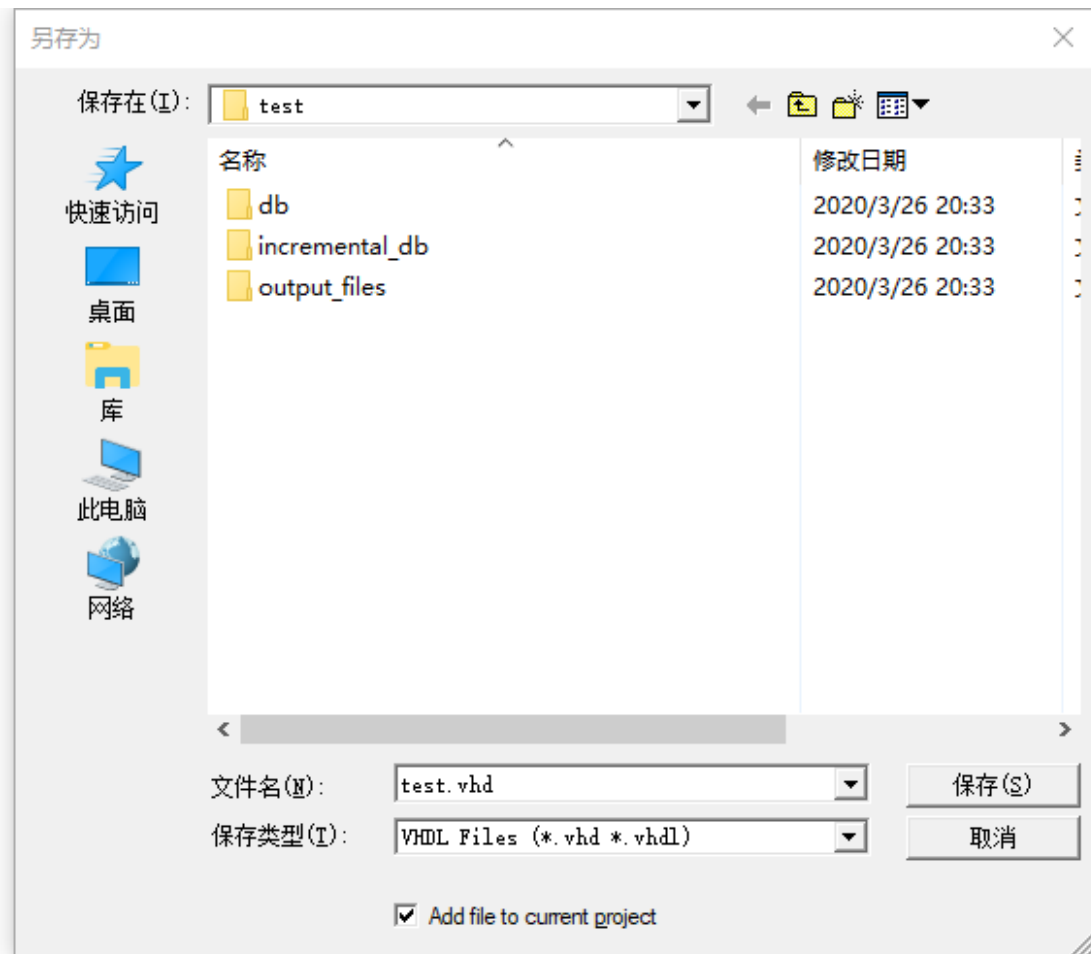
1. 在 File 选项下，选择 New，选择 VHDL File 点击 ok



2. 写入代码，此处详见课上内容，不做说明

```
library ieee;
use ieee.std_logic_1164.all;
entity Zuo_Dcf is
    port(Re_L,d,cp : in std_logic;nq : out std_logic;q : buffer std_logic);
end Zuo_Dcf;
architecture do of Zuo_Dcf is
    begin
        process(Re_L,cp)
        begin
            if(Re_L='0')
            then q<='0';
            elsif(cp'event and cp='1')
            then q<=d;
            end if;
            nq<=not q;
        end process;
    end do;
```

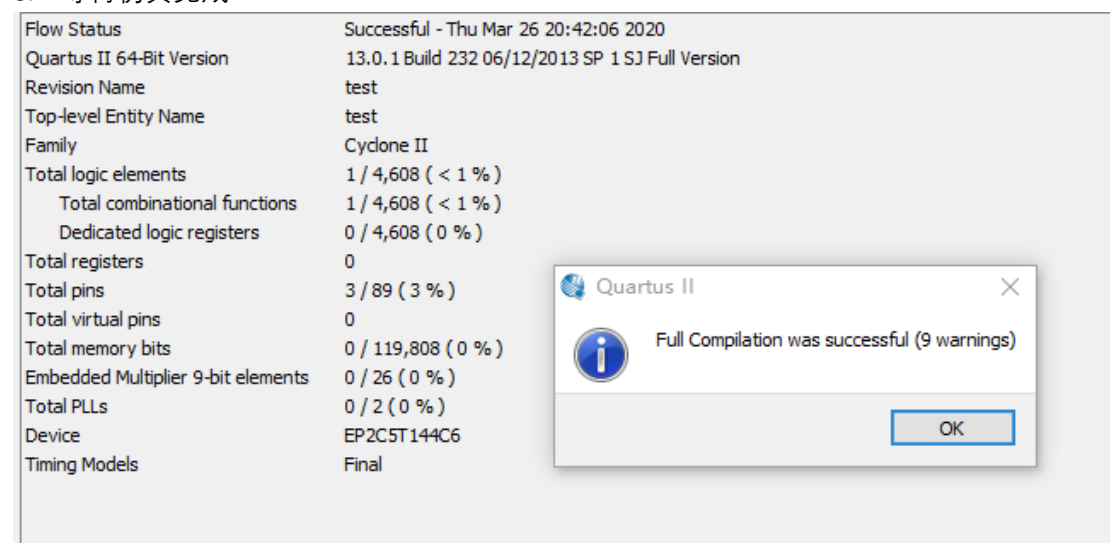
3. 保存文件，单个文件时文件名与顶层实体名相同，文件扩展名为.vhd，选择保存



4. 运行仿真，找到工具栏中红圈按钮（或者红圈后面的按钮），并点击

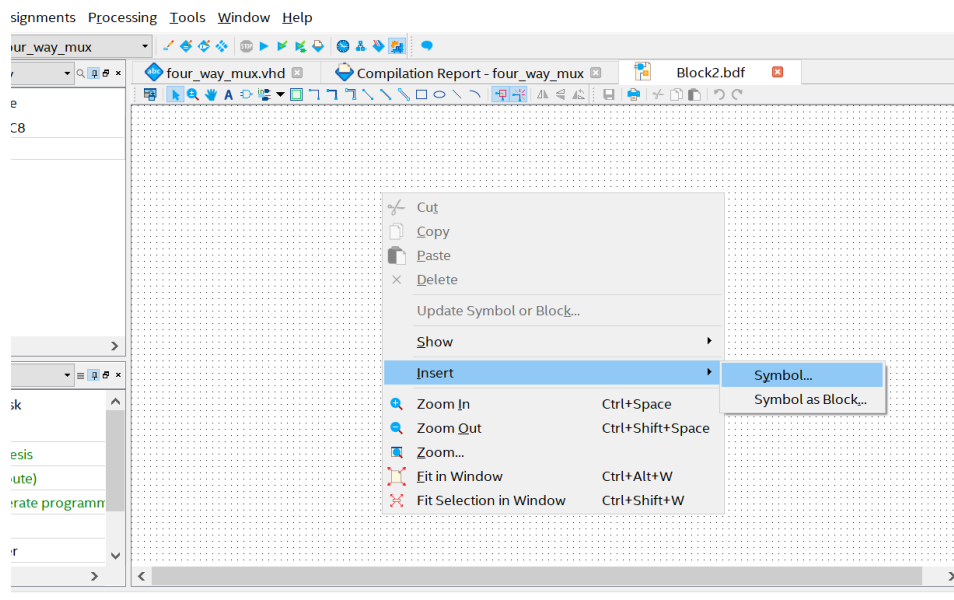
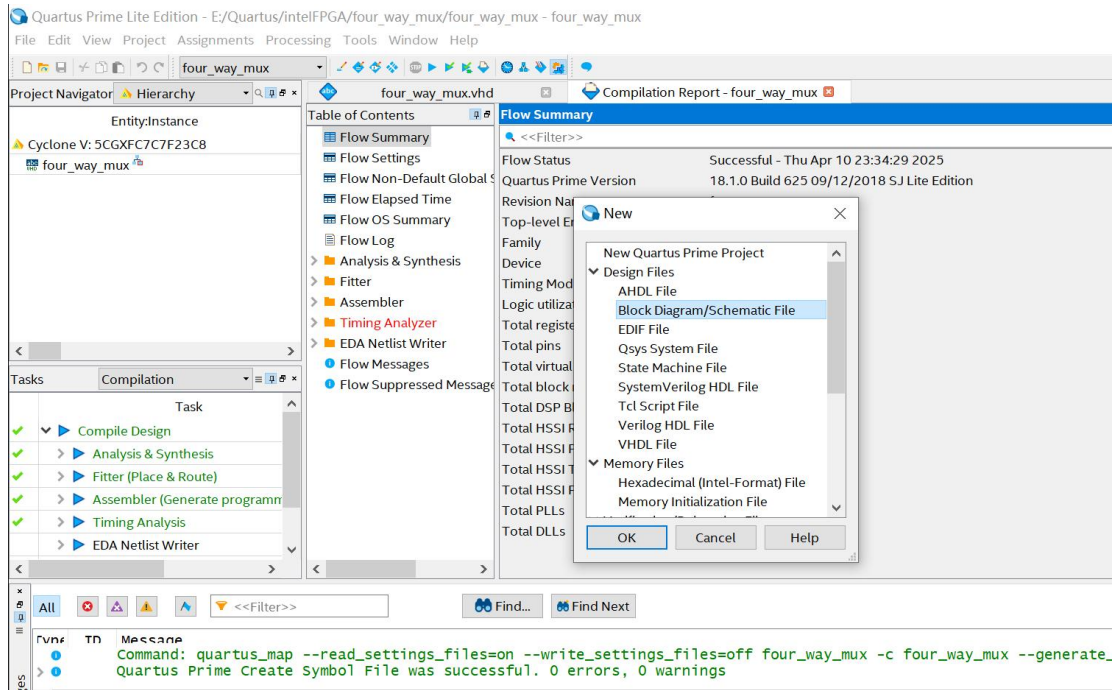


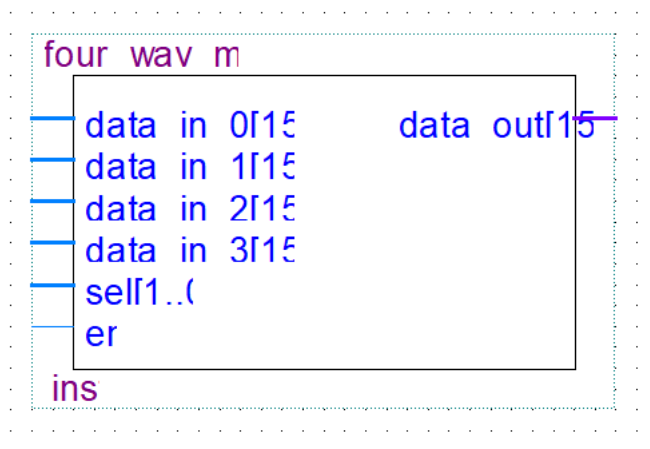
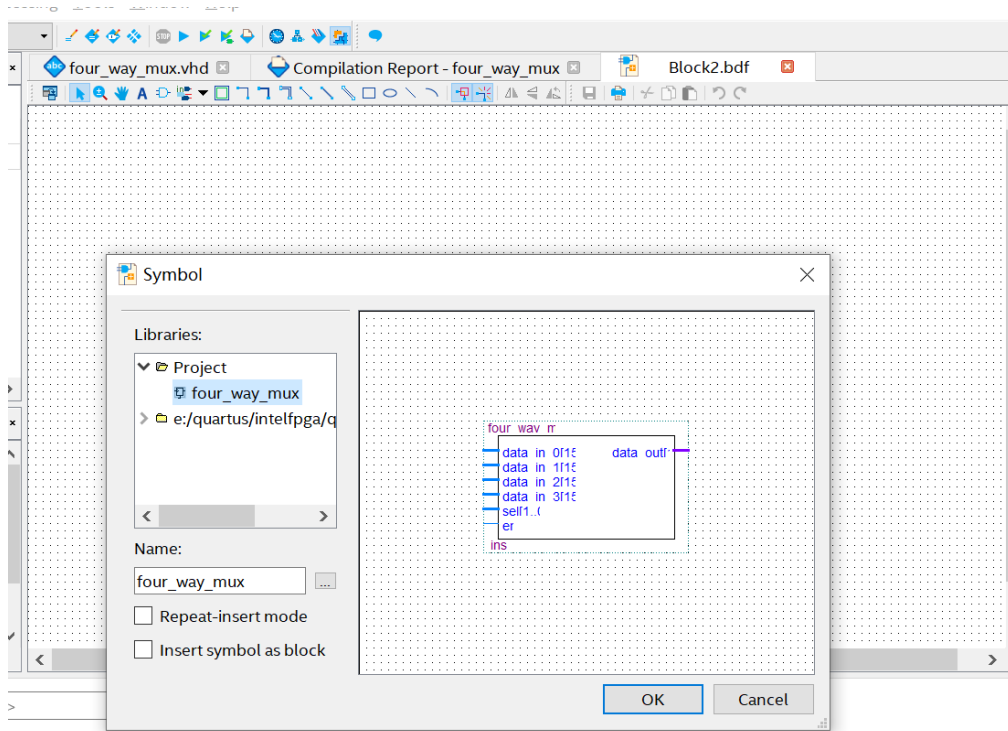
5. 等待仿真完成



(1) 创建逻辑符号

选择“File”->“Create/Update”->“Create Symbol Files for Current File”，

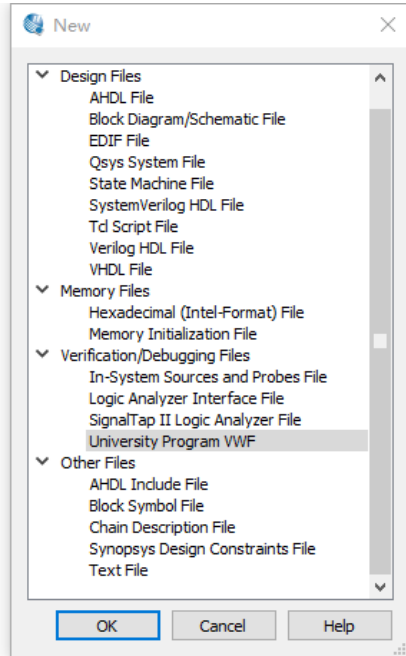




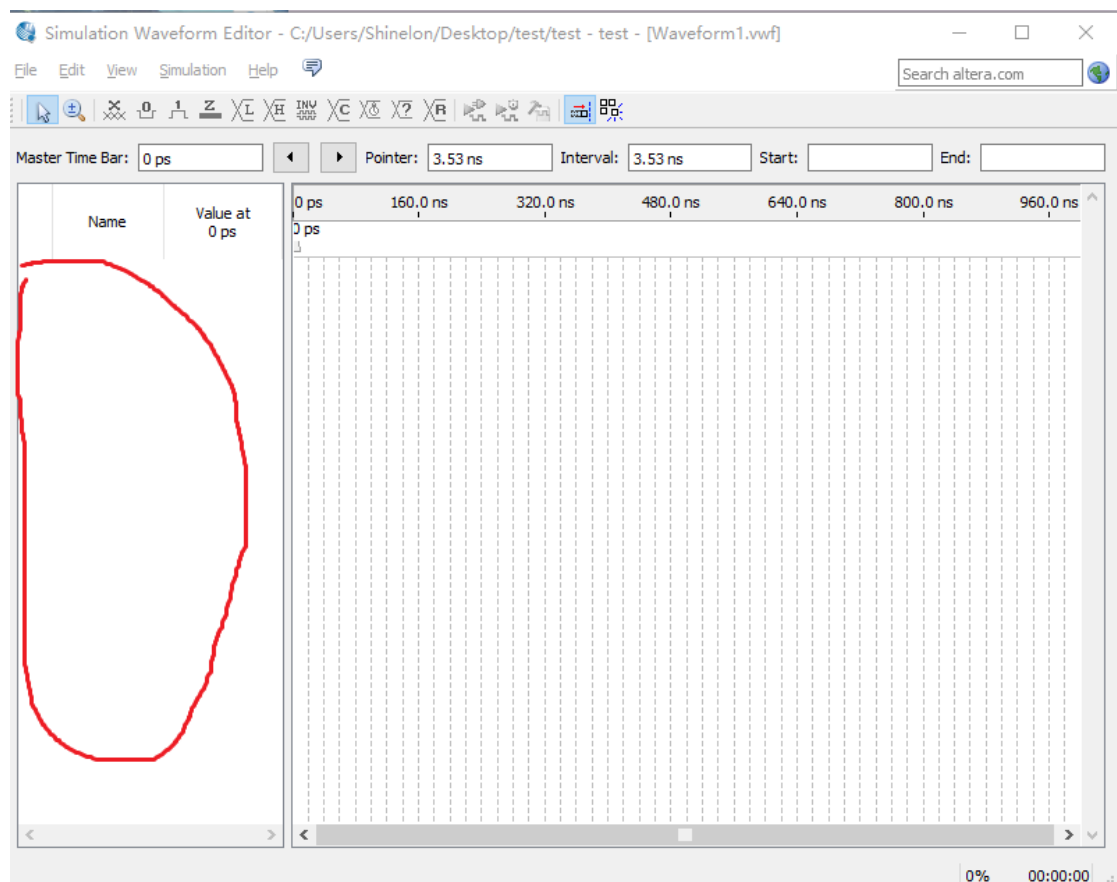
波形图文件的创建以及仿真

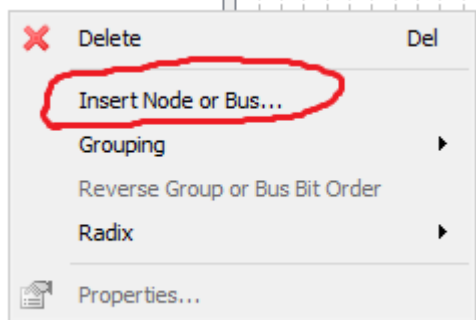
注：本次仿真为二输入与门

1. 在 File 选项下，选择 New，选择 University Program VMF，点击 ok

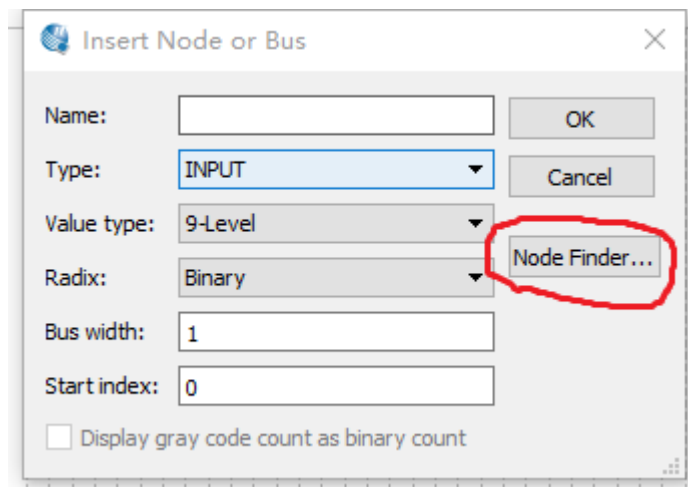


2. 在红色圈内右键，选择 Insert Node or Bus

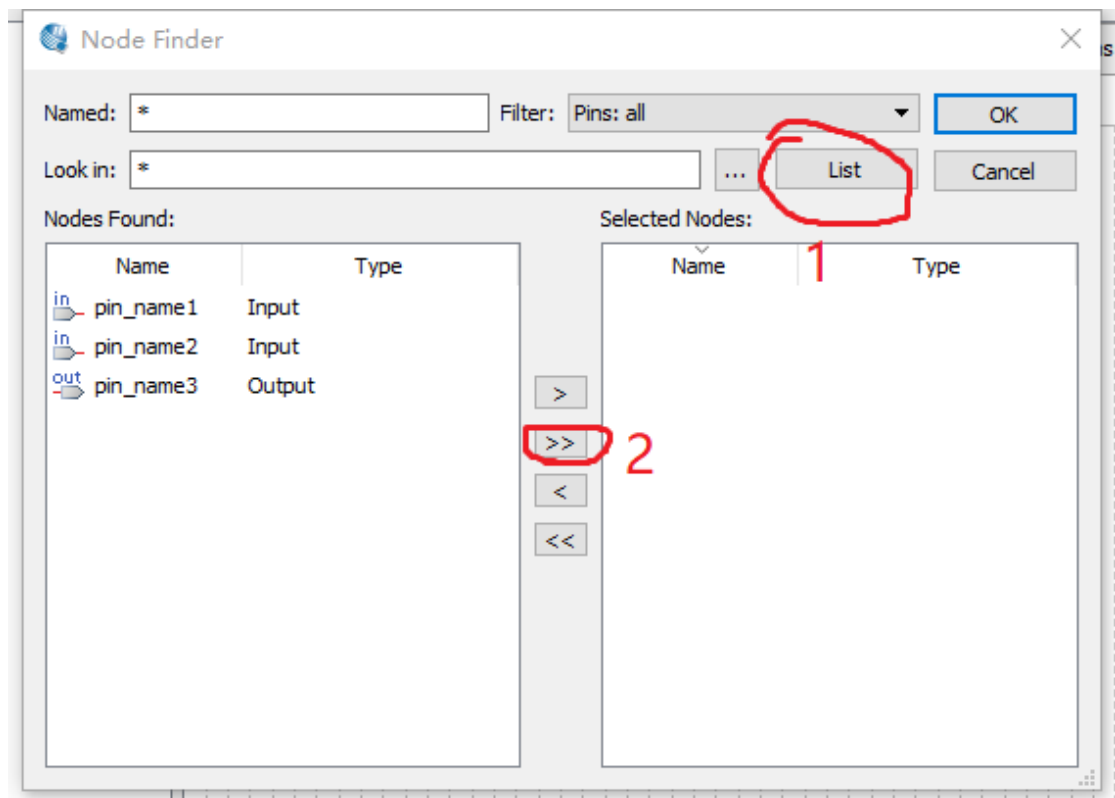




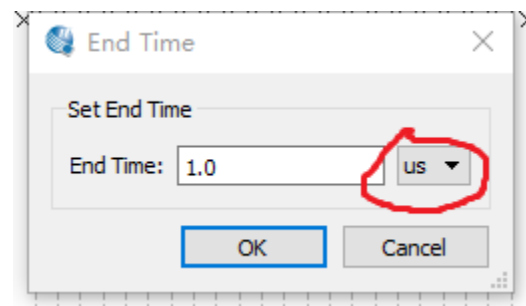
3. 在弹出的窗口内点击 Node Finder



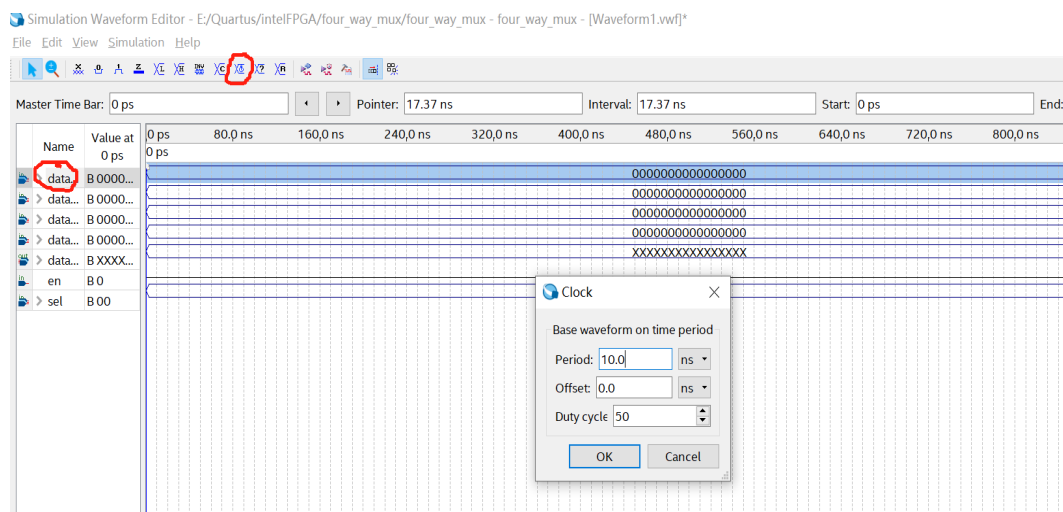
4. 点击 List 按钮，然后点击>>将引脚全部选中（或者选择部分引脚，点击>），点击 ok



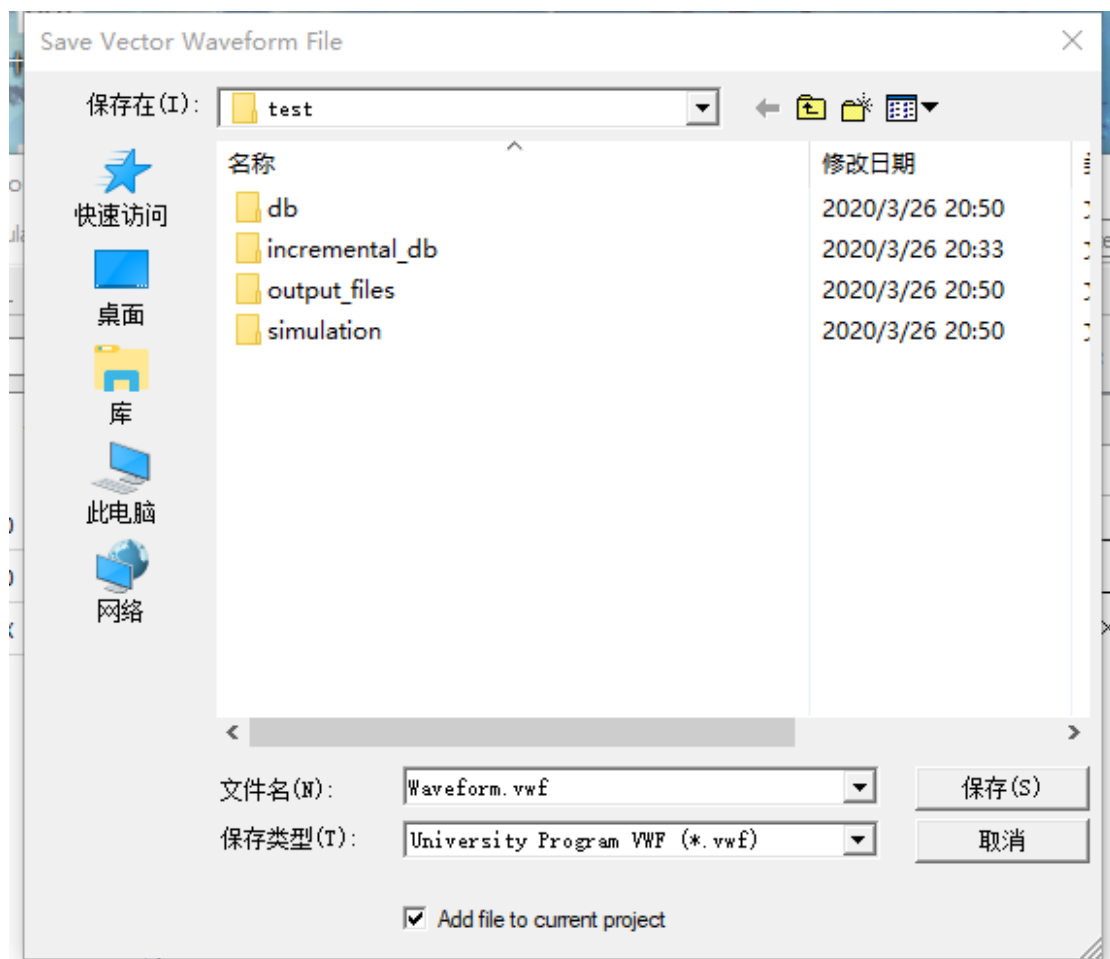
5. 在 Edit 选项下，选择 Set End Time,在弹出的窗口自定义仿真时间，本实例为 20ns
注：注意时间单位



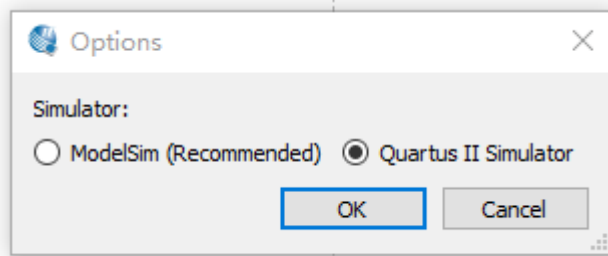
6. 对波形图进行设置（调整脉冲周期）



7. 保存文件，文件名自设，文件扩展名为.vmf，选择保存



8. 在 Simulation 选项下，选择 Options，在弹出的窗口中选择 Quartus II Simulator，点击 ok



9. 点击工具栏中红圈内所示的按钮



10. 等待仿真结束以及仿真结果

